

THE SCOTS COLLEGE



TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

1999

3/4 UNIT MATHEMATICS

TIME ALLOWED: Two Hours
 [plus 5 minutes reading time]

INSTRUCTIONS TO CANDIDATES:

- ALL QUESTIONS ARE TO BE ATTEMPTED.
- ALL QUESTIONS ARE OF EQUAL VALUE.
- ALL NECESSARY WORKING SHOULD BE SHOWN FOR EACH QUESTION.
- NON-PROGRAMMABLE CALCULATORS ARE PERMITTED.
- A TABLE OF STANDARD INTEGRALS IS PROVIDED.

BOOKLET ORDER:

BOOKLET 1 : QUESTIONS 1 & 2
BOOKLET 2 : QUESTIONS 3 & 4
BOOKLET 3 : QUESTIONS 5, 6 & 7

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Exam continues over

QUESTION 1

- (a) Differentiate $e^x \cos(e^{-x})$. [2 MARKS]
- (b) If $\log_2 3 = b$ write $\log_4 27$ in terms of b . [1 MARK]
- (c) If u, v and w are the roots of $x^3 - 4x + 1 = 0$, find the value of $\frac{1}{u} + \frac{1}{v} + \frac{1}{w}$. [2 MARKS]
- (d) Find the exact value of $\tan^{-1} \sqrt{3} - \tan^{-1}(-1)$. [2 MARKS]
- (e) Solve the inequality $\frac{5}{2x-1} \geq 1$. [3 MARKS]
- (f) Find the distance between $y = 2x + 1$ and $2x - y + 8 = 0$. [2 MARKS]

QUESTION 2

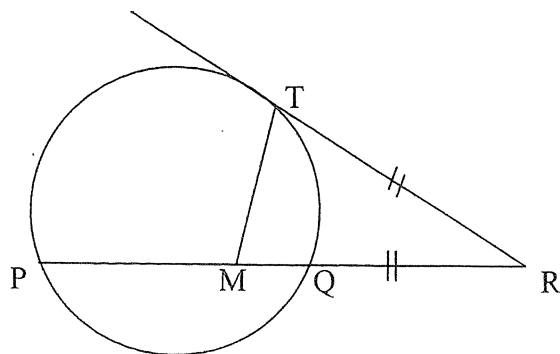
- (a) Find the primitive of $\sin^2 \frac{x}{2}$. [2 MARKS]
- (b) Find the value of $\int_{\frac{\pi}{8}}^{\frac{7\pi}{6}} \tan 2x \sec^2 2x$ using the substitution $u = \tan 2x$. [3 MARKS]
- (c) $P(2ap, ap^2)$ is a point on the parabola $x^2 = 4ay$. The tangent at P cuts the axis of the parabola at M and G is the foot of the perpendicular from P to the axis of the parabola. Prove that M and G are equidistant from the vertex for all positions of P . [4 MARKS]
- (d) Solve for $0^\circ \leq x \leq 360^\circ$
 $\sin x + \cos x = \frac{4}{5}$. [3 MARKS]

QUESTION 3

- (a) A particle moves along the x axis with acceleration $35 + 6t - 6t^2$. If the particle is initially at rest at the origin, find its maximum displacement in the positive direction.

[5 MARKS]

(b)



In the diagram RT is a tangent to the circle and RQP is a secant cutting the circle in Q and P. M is a point on PQ so that RM = RT.

Copy the diagram and prove that MT bisects the angle PTQ.

[4 MARKS]

- (c) Prove by induction that $7^n + 5$ is divisible by 3 where n is any positive integer.

[3 MARKS]

QUESTION 4

- (a) Prove for any acute angle θ that $\cot \theta = \sqrt{\frac{1 + \cos 2\theta}{1 - \cos 2\theta}}$. Hence show that the exact

value of $\cot \frac{\pi}{8}$ is $\sqrt{2} + 1$.

[4 MARKS]

- (b) Two sides of a triangle are 10cm and 18cm respectively. If the angle between them is increasing at the rate of $\frac{1}{10}$ radian per day, find how fast the area of the triangle

is increasing when the angle is $\frac{\pi}{3}$.

[3 MARKS]

- (c) The rate of growth of the number of goats on a farm is given by $\frac{dN}{dt} = k(N - 200)$

where N is the number of goats.

[5 MARKS]

(i) Show that $N = 200 + Ae^{kt}$ is a solution of the differential equation.

(ii) If initially there were 300 goats and 2 years later there were 420, find the number of goats on the farm at the end of 4 years.

[START A NEW BOOKLET]

QUESTION 5

- (a) State the domain and range of the function $y = 4 \sin^{-1} 2x$. Sketch this function. [4 MARKS]
- (b) Show that $\frac{d}{dx}(x \tan^{-1} x) = \frac{x}{1+x^2} + \tan^{-1} x$. Hence evaluate $\int_0^1 \tan^{-1} x \, dx$, giving your answer correct to 3 significant figures. [5 MARKS]
- (c) If $y = x \tan^{-1} x - \log_e \sqrt{1+x^2}$ show that $(x^2 + 1) \frac{d^2 y}{dx^2} = 1$. [3 MARKS]

QUESTION 6

- (a) A particle is projected with initial velocity 40 m/sec in a direction making 30° with the horizontal. Using $g = 9.8 \text{ m/s}^2$, find: [7 MARKS]
- (i) the height of the particle after 3 seconds;
- (ii) the cartesian equation of its path.
- (b) A particle moves so that its distance x cm from a fixed point O after t seconds is given by $x = 4 \cos 3t$. [5 MARKS]
- (i) Show that the particle is moving in Simple Harmonic Motion.
- (ii) What is the period of the motion?
- (iii) Find the speed when the particle is first 1 cm from O.

QUESTION 7

- (a) Find the area bounded by the curve $y = \frac{1}{\sqrt{16-x^2}}$, the x axis and the ordinates $x = -2$ and $x = 2$. [3 MARKS]
- (b) Show that $f(x) = 4 - 10x + 3 \sin x$ has a zero between $x = 0$ and $x = 1$. Use $x = 0.5$ and one application of Newton's method to find a better approximation. (Answer correct to 3 dec. pl) [5 MARKS]
- (c) Find the volume of the solid formed by rotating $\frac{x^2}{9} + \frac{y^2}{4} = 1$ about the x axis. [4 MARKS]

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0.$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0.$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0.$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0.$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, a \neq 0.$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0.$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0.$$

$$\int \frac{1}{\sqrt{(a^2 - x^2)}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a.$$

$$\int \frac{1}{\sqrt{(x^2 - a^2)}} dx = \ln \left\{ x + \sqrt{(x^2 - a^2)} \right\}, |x| > |a|$$

$$\int \frac{1}{\sqrt{(x^2 + a^2)}} dx = \ln \left\{ x + \sqrt{(x^2 + a^2)} \right\}$$

NOTE: $\ln x = \log_e x, x > 0$

THE SCOTS COLLEGE 1999 3 UNIT TRIAL

1 (a) Let $y = e^x \cos e^{-x}$
 $\frac{dy}{dx} = e^x \cos e^{-x} + e^x \cdot (-e^{-x}) \cdot \sin e^{-x}$
 $= e^x \cos e^{-x} - \sin e^{-x}$

(b) $\log_4 27 = \frac{\log 27}{\log 4}$
 $= \frac{3 \log 3}{2 \log 2}$
 $= \frac{3 \log 3}{2}$
 $= \frac{3 \log 3}{2}$

(c) $\frac{1}{u} + \frac{1}{v} + \frac{1}{w}$
 $= \frac{vw + uw + uv}{uvw}$
 $= \frac{-4}{-1}$
 $= 4$

(d) $\tan^{-1} \sqrt{3} - \tan^{-1}(-1)$
 $= \frac{\pi}{3} - (-\frac{\pi}{4})$
 $= \frac{7\pi}{12}$

(e) $\frac{5}{2x-1} \geq 1 \quad x \neq \frac{1}{2}$
 $(2x-1)^2 \frac{5}{2x-1} \geq (2x-1)^2$
 $10x - 5 \geq 4x^2 - 4x + 1$
 $4x^2 - 14x + 6 \leq 0$
 $2x^2 - 7x + 3 \leq 0$
 $(2x-1)(x-3) \leq 0$
 $x \neq \frac{1}{2}, \frac{1}{2} < x \leq 3$

(f) $y = 2x + 1$ $(0, 1)$ is on line
 $p = \frac{0 - 1 + 8}{\sqrt{5}} = \frac{7}{\sqrt{5}} = \frac{7\sqrt{5}}{5}$

2 (a) $\int \sin^2 \frac{x}{2} dx$
 $= \int \frac{1 - \cos x}{2} dx$
 $= \frac{x}{2} - \frac{\sin x}{2} + C$

(b) $u = \tan 2x$
 $du = 2 \sec^2 2x dx$
 when $x = \frac{7\pi}{6}$, $u = \sqrt{3}$
 when $x = \frac{\pi}{8}$, $u = 1$
 $\int_{\frac{\pi}{8}}^{\frac{7\pi}{6}} \tan 2x \sec^2 2x dx$
 $= \int_1^{\sqrt{3}} u \cdot \frac{1}{2} du$
 $= \left[\frac{u^2}{4} \right]_1^{\sqrt{3}}$
 $= \frac{3}{4} - \frac{1}{4}$
 $= \frac{1}{2}$

(c) $x^2 = 4ay$

 Gradient of tangent = p
 Tangent at P is
 $y - ap^2 = p(x - xp)$
 $y = px - ap^2$
 when $x = 0$ (axis of parabola)
 $y = -ap^2$
 $\therefore M$ is $(0, -ap^2)$
 G is $(0, ap^2)$

$OG = ap^2$
 $OM = ap^2$
 $\therefore OG = OM$

$$2(a) \sin x + \cos x = \frac{4}{5}$$

$$\frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2} = \frac{4}{5} \quad (t = \tan \frac{x}{2})$$

$$10t + 5 - 5t^2 = 4 + 4t^2$$

$$9t^2 - 10t - 1 = 0$$

$$t = \frac{10 \pm \sqrt{136}}{18}$$

$$t = -0.0923 \text{ or } t = 1.2034 \quad (0 \leq \frac{x}{2} \leq 180^\circ)$$

$$\frac{x}{2} = 174^\circ 44' \text{ or } \frac{x}{2} = 50^\circ 17'$$

$$x = 349^\circ 34' \text{ or } 100^\circ 34'$$

3.

$$(a) a = 35 + 6t - 6t^2$$

$$\frac{dv}{dt} = 35 + 6t - 6t^2$$

$$v = 35t + 3t^2 - 2t^3 + C$$

$$v = 0 \text{ when } t = 0 \therefore C = 0$$

$$v = 35t + 3t^2 - 2t^3$$

max. displacement when $v = 0$

$$t(35 + 3t - 2t^2) = 0$$

$$t(7 + 2t)(5 - t) = 0$$

$$t = 0 \text{ or } t = 5 \quad (t \neq -\frac{7}{2})$$

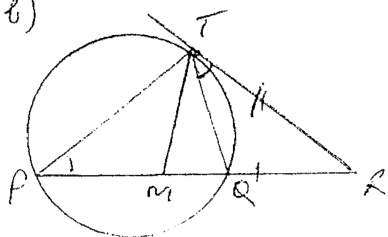
$$\text{displacement } x = \frac{35t^2}{2} + t^3 - \frac{t^4}{2} + C$$

$$x = 0 \text{ when } t = 0 \therefore C = 0$$

$$x = \frac{35t^2}{2} + t^3 - \frac{t^4}{2}$$

$$t = 5, \quad x = \frac{35(25)}{2} + 125 - \frac{625}{2}$$

(b)



$$TR = MR \therefore \angle RTM = \angle RMT \text{ (base } \angle \text{ of } \triangle TRM)$$

$$\angle RTQ = \angle TPQ \text{ (Tangent/angle segment)}$$

By subtraction,

$$\angle MTR = \angle RMT - \angle TPQ$$

$$= \angle RMT \text{ (ext } \angle \text{ sum of } \triangle TPQ)$$

$$\therefore MT \text{ bisects } \angle PRQ$$

(c)

$$(i) n = 1$$

$$7+5=12 \text{ (divisible by 3)}$$

(ii) assume true for $n = k$

$$\text{let } 7^k + 5 = 3m \text{ (m an integ)}$$

(iii) when $n = k+1$

$$7^{k+1} + 5 = 7 \cdot 7^k + 5$$

$$= 7(3m - 5) + 5$$

$$= 21m - 30$$

$\therefore$ divisible by 3

It is true for $n = 1$ and

therefore $n = 2, n = 3$ and so on

Hence true for all n .

4.

$$(a) \frac{1 + \cos 2\theta}{1 - \cos 2\theta} = \frac{1 + 2\cos^2 \theta - 1}{1 - (1 - 2\sin^2 \theta)}$$

$$= \frac{2\cos^2 \theta}{2\sin^2 \theta}$$

$$= \cot^2 \theta$$

$$\therefore \sqrt{\frac{1 + \cos 2\theta}{1 - \cos 2\theta}} = \cot \theta$$

$$\cot \frac{\pi}{8} = \sqrt{\frac{1 + \cos \frac{\pi}{4}}{1 - \cos \frac{\pi}{4}}}$$

$$= \sqrt{\frac{1 + \frac{\sqrt{2}}{2}}{1 - \frac{\sqrt{2}}{2}}}$$

$$= \sqrt{\frac{2 + \sqrt{2}}{2 - \sqrt{2}}}$$

$$= \sqrt{\frac{(2 + \sqrt{2})(2 + \sqrt{2})}{(2 - \sqrt{2})(2 + \sqrt{2})}}$$

$$= \frac{2 + \sqrt{2}}{\sqrt{2}}$$

$$= 1 + \sqrt{2}$$

$$(b) A = \frac{1}{2} ab \sin \theta$$

$$= 90 \sin \theta$$

$$\frac{dA}{d\theta} = 90 \cos \theta = 45 \text{ when } \theta = \frac{\pi}{3}$$

$$\frac{dA}{dt} = \frac{dA}{d\theta} \cdot \frac{d\theta}{dt}$$

$$= 45 \cdot \frac{1}{10}$$

$$= 4.5 \text{ cm}^2/\text{day}$$

(c)

$$\frac{dN}{dt} = k(N-200)$$

$$(i) N = 200 + Ae^{kt}$$

$$\frac{dN}{dt} = kAe^{kt}$$

$$= k(N-200)$$

$\therefore N = 200 + Ae^{kt}$ is a solution.

$$(ii) N = 200 + Ae^{kt}$$

$$t=0, N=300$$

$$300 = 200 + Ae^0$$

$$A = 100$$

$$\therefore N = 200 + 100e^{kt}$$

$$t=2, N=420$$

$$420 = 200 + 100e^{2k}$$

$$220 = 100e^{2k}$$

$$e^{2k} = 2.2$$

$$k = \frac{\ln 2.2}{2}$$

$$k = 0.39442$$

$$\therefore N = 200 + 100e^{0.3944t}$$

$$\text{When } t=4, N = 200 + 100e^{4(0.3944)}$$

$$= 684 \text{ goats}$$

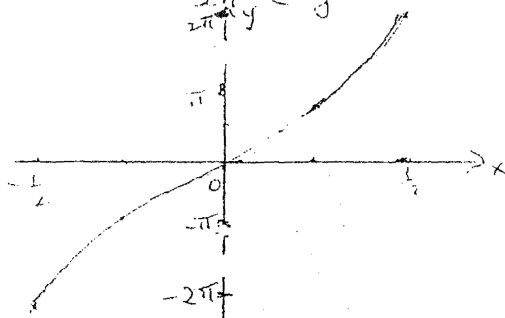
$$5(a) y = 4 \sin^{-1} 2x$$

$$\text{domain } -1 \leq 2x \leq 1$$

$$-\frac{1}{2} \leq x \leq \frac{1}{2}$$

$$\text{range } 4\left(-\frac{\pi}{2}\right) \leq y \leq 4\left(\frac{\pi}{2}\right)$$

$$-2\pi \leq y \leq 2\pi$$



$$(b) \frac{d}{dx}(x \tan^{-1} x) = \tan^{-1} x + x \cdot \frac{1}{1+x^2}$$

$$= \tan^{-1} x + \frac{x}{1+x^2}$$

$$\therefore \int \left(\tan^{-1} x + \frac{x}{1+x^2} \right) dx = x \tan^{-1} x$$

$$\int \tan^{-1} x dx = x \tan^{-1} x - \int \frac{x}{1+x^2} dx$$

$$\int_0^1 \tan^{-1} x dx = \left[x \tan^{-1} x \right]_0^1 - \int_0^1 \frac{x}{1+x^2} dx$$

$$= \frac{\pi}{4} - \frac{1}{2} \left[\ln(1+x^2) \right]_0^1$$

$$= \frac{\pi}{4} - \frac{1}{2} (\ln 2 - 0)$$

$$= \frac{\pi}{4} - \frac{1}{2} \ln 2$$

$$= 0.439 \text{ (3 sig. fig.)}$$

$$(c) y = x \tan^{-1} x - \ln \sqrt{1+x^2}$$

$$= x \tan^{-1} x - \frac{1}{2} \ln(1+x^2)$$

$$\frac{dy}{dx} = \tan^{-1} x + x \cdot \frac{1}{1+x^2} - \frac{2x}{2(1+x^2)}$$

$$= \tan^{-1} x + \frac{x}{1+x^2} - \frac{x}{1+x^2}$$

$$= \tan^{-1} x$$

$$\frac{d^2y}{dx^2} = \frac{1}{1+x^2}$$

$$(x^2+1) \frac{d^2y}{dx^2} = x^2+1 \cdot \frac{1}{1+x^2} = 1$$

Horizontally $\dot{x} = V \cos \alpha$
 $= 40 \cos 30^\circ$
 $= 20\sqrt{3}$

Vertically $\dot{y} = V \sin \alpha - gt$
 $= 40 \sin 30^\circ - 9.8t$
 $= 20 - 9.8t$

Distance $x = V \cos \alpha t$ — (1)
 $= 20\sqrt{3} \cdot 3$
 $= 60\sqrt{3}$

$y = V \sin \alpha t - \frac{1}{2}gt^2 + c$

$t=0, y=0$

$y = V \sin \alpha t - \frac{1}{2}gt^2$

$y = 20t - 4.9t^2$ — (2)

$t=3, y = 15.9 \text{ m.}$

From (1) $t = \frac{x}{20\sqrt{3}}$

In (2)

$y = \frac{20x}{20\sqrt{3}} - 4.9 \left(\frac{x}{20\sqrt{3}} \right)^2$

$y = \frac{x}{\sqrt{3}} - \frac{4.9x^2}{1200}$

(8) (i) $\dot{x} = 4 \cos 3t$
 $\ddot{x} = -12 \sin 3t$
 $\ddot{x} = -36 \cos 3t$
 $= -9x$

In form $\ddot{x} = -n^2x$ where $n=3$

S.H.M.

(ii) $T = \frac{2\pi}{n} = \frac{2\pi}{3}$
 period

(iii) when $x=1$

$1 = 4 \cos 3t$

$\cos 3t = \frac{1}{4}$

$3t = 1.32$

$t = 0.44$

$\dot{x} = -12 \sin 3t$
 $= -12 \sin 1.32$
 $= -11.6 \text{ m/s}$

7 (a) $A = \int y dx$
 $= \int_{-2}^2 \frac{1}{\sqrt{16-x^2}} dx$
 $= \left[\sin^{-1} \frac{x}{4} \right]_{-2}^2$
 $= \sin^{-1} \frac{1}{2} - \sin^{-1} \left(-\frac{1}{2} \right)$
 $= \frac{\pi}{6} - \left(-\frac{\pi}{6} \right) = \frac{\pi}{3} \text{ m}^2$

(b) $x=0, f(0) = 4 - 0 + 0 = 4$
 $x=1, f(1) = 4 - 10 + 3 = -3$

change of sign
 $\therefore 0 < x < 1$

$f(0.5) = 0.4383 = f(a)$

$f'(x) = -10 + 3 \cos x = f'(a)$

$f'(0.5) = -7.3673$

$x_1 = a - \frac{f(a)}{f'(a)}$

$= 0.5 - \frac{0.4383}{-7.3673}$

$= 0.55949$

$= 0.559 \text{ (3 dec. pl.)}$

(c) $\frac{x^2}{9} + \frac{y^2}{4} = 1$ $\frac{y^2}{4} = 1 - \frac{x^2}{9}$

$V = \pi \int y^2 dx$ when $y=0$
 $x = \pm 3$

$= \pi \int_{-3}^3 \left(4 - \frac{4x^2}{9} \right) dx$

$= \pi \left[4x - \frac{4x^3}{27} \right]_{-3}^3$

$= \pi \left[(12 - 4) - (-12 + 4) \right]$

$= 16\pi \text{ m}^3$